

Практическое занятие 10

Правило Лопиталя.

Пользуясь правилом Лопиталя, раскрыть неопределенности $\left(\frac{0}{0}\right)$ или $\left(\frac{\infty}{\infty}\right)$

$$1) \lim_{x \rightarrow 1} \frac{\sqrt[7]{4x-3} - \sqrt[5]{3x-2}}{\ln(7-6x)} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 1} \frac{\frac{4}{7\sqrt[7]{(4x-3)^6}} - \frac{3}{5\sqrt[5]{(3x-2)^4}}}{\frac{-6}{7-6x}} = \frac{\frac{4}{7} - \frac{3}{5}}{-6} = \frac{1}{210}$$

$$2) \lim_{x \rightarrow 0+0} (x \ln x) = (0 * \infty) = \lim_{x \rightarrow 0+0} \frac{\ln x}{\frac{1}{x}} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow 0+0} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0+0} (-x) = 0$$

$$3) \lim_{x \rightarrow 0+0} \frac{\ln(\sin 3x)}{\ln(\sin 7x)} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow 0+0} \frac{\frac{1}{\sin 3x} \cos 3x * 3}{\frac{1}{\sin 7x} \cos 7x * 7} = \lim_{x \rightarrow 0+0} \frac{3 \cos 3x \sin 7x}{7 \cos 7x \sin 3x} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0+0} \frac{3 * 7x \cos 3x}{7 * 3x \cos 7x} = 1$$

$$4) \lim_{x \rightarrow 0+0} \frac{\ln(\arcsin 3x)}{\ln(\arcsin 7x)} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow 0+0} \frac{\frac{1}{\arcsin 3x} * \frac{3}{\sqrt{1-9x^2}}}{\frac{1}{\arcsin 7x} * \frac{7}{\sqrt{1-49x^2}}} = \lim_{x \rightarrow 0+0} \frac{3 \sqrt{1-49x^2} \arcsin 7x}{7 \sqrt{1-9x^2} \arcsin 3x} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0+0} \frac{3 * 7x \sqrt{1-49x^2}}{7 * 3x \sqrt{1-9x^2}} = 1$$

$$5) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\ln(\operatorname{tg} 5x)}{\pi - 4x} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\frac{1}{\operatorname{tg} 5x} * \frac{1}{\cos^2 5x} 5}{-4} = \frac{1 * 2 * 5}{-4} = -\frac{5}{2}$$

$$6) \lim_{x \rightarrow \frac{3\pi}{2}} \frac{\sin 4x}{\cos 3x} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow \frac{3\pi}{2}} \frac{4 \cos 4x}{-3 \sin 3x} = \frac{4 * 1}{-3 * 1} = -\frac{4}{3}$$

$$7) \lim_{x \rightarrow 0+0} (x^2 \ln 2x) = (0 * \infty) = \lim_{x \rightarrow 0+0} \frac{\ln 2x}{\frac{1}{x^2}} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow 0+0} \frac{\frac{1}{2x} 2}{-\frac{2}{x^3}} = \lim_{x \rightarrow 0+0} \left(-\frac{x^2}{2}\right) = 0$$

$$8) \lim_{x \rightarrow +\infty} \frac{\ln 2x}{\sqrt{x}} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow +\infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow +\infty} \left(\frac{2}{\sqrt{x}}\right) = 0$$

$$9) \lim_{x \rightarrow +\infty} (x e^{-2x}) = (\infty * 0) = \lim_{x \rightarrow +\infty} \frac{x}{e^{2x}} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow +\infty} \frac{1}{2 e^{2x}} = 0$$

$$10) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt[5]{\operatorname{tg} x} - 1}{\cos 2x} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\frac{1}{5\sqrt[5]{\operatorname{tg}^4 x}} * \frac{1}{\cos^2 x}}{-2 \sin 2x} = -\frac{1}{5}$$

$$11) \lim_{x \rightarrow 1} \frac{\ln(3x-2)}{\sqrt{x} - \sqrt[3]{x}} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 1} \frac{\frac{3}{3x-2}}{\frac{1}{2\sqrt{x}} - \frac{1}{3\sqrt[3]{x^2}}} = \frac{3}{\frac{1}{2} - \frac{1}{3}} = 18$$

$$12) \lim_{x \rightarrow +\infty} \frac{\ln(2+3^x)}{\ln(3+2^x)} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow +\infty} \frac{\frac{3^x \ln 3}{2+3^x}}{\frac{2^x \ln 2}{3+2^x}} = \lim_{x \rightarrow +\infty} \frac{(3+2^x) 3^x \ln 3}{(2+3^x) 2^x \ln 2} = \lim_{x \rightarrow +\infty} \frac{\left(\frac{3}{2^x} + 1\right) \ln 3}{\left(\frac{2}{3^x} + 1\right) \ln 2} = \frac{\ln 3}{\ln 2}$$

Повторное применение правила Лопиталя

$$13) \quad \lim_{x \rightarrow 0} \frac{e^x - \cos x - x}{x^2} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{2x} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{e^x + \cos x}{2} = 1$$

$$14) \quad \lim_{x \rightarrow 0} \frac{\ln(1+x) - \sin x}{x^2} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{\frac{1}{1+x} - \cos x}{2x} = \lim_{x \rightarrow 0} \frac{1 - \cos x - x \cos x}{2x + 2x^2} = \left(\frac{0}{0}\right) =$$

$$\lim_{x \rightarrow 0} \frac{\sin x - \cos x + x \sin x}{2 + 4x} = -\frac{1}{2}$$

$$15) \quad \lim_{x \rightarrow +\infty} (x^2 e^{-3x}) = (\infty * 0) = \lim_{x \rightarrow +\infty} \frac{x^2}{e^{3x}} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow +\infty} \frac{2x}{3e^{3x}} = \left(\frac{\infty}{\infty}\right) =$$

$$\lim_{x \rightarrow +\infty} \frac{2}{9e^{3x}} = 0$$

$$16) \quad \lim_{x \rightarrow 0+0} (\sqrt{x} \ln^2 x) = (0 * \infty) = \lim_{x \rightarrow 0+0} \frac{\ln^2 x}{\frac{1}{\sqrt{x}}} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow 0+0} \frac{2(\ln x) * \frac{1}{x}}{-\frac{1}{2x\sqrt{x}}} =$$

$$\lim_{x \rightarrow 0+0} \left(-\frac{4 \ln x}{\frac{1}{\sqrt{x}}}\right) = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow 0+0} \left(-\frac{\frac{4}{x}}{-\frac{1}{2x\sqrt{x}}}\right) = \lim_{x \rightarrow 0+0} (8\sqrt{x}) = 0$$

$$17) \quad \lim_{x \rightarrow +\infty} \frac{\ln(1+4x^2)}{\ln\left(\frac{\pi}{2} - \arctg 2x\right)} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow +\infty} \frac{\frac{8x}{1+4x^2}}{\frac{1}{\frac{\pi}{2} - \arctg 2x} \left(-\frac{2}{1+4x^2}\right)} =$$

$$\lim_{x \rightarrow +\infty} \left(-4x \left(\frac{\pi}{2} - \arctg 2x\right)\right) = (\infty * 0) = \lim_{x \rightarrow +\infty} \frac{\left(-4\left(\frac{\pi}{2} - \arctg 2x\right)\right)}{\frac{1}{x}} = \left(\frac{0}{0}\right) =$$

$$\lim_{x \rightarrow +\infty} \frac{4 \frac{2}{1+4x^2}}{-\frac{1}{x^2}} = \lim_{x \rightarrow +\infty} \frac{-8x^2}{1+4x^2} = -2$$

$$18) \quad \lim_{x \rightarrow 0} \frac{\arcsin 3x - 3 \arcsin x}{x^3} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{\frac{3}{\sqrt{1-9x^2}} - \frac{3}{\sqrt{1-x^2}}}{3x^2} =$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1-x^2} - \sqrt{1-9x^2}}{x^2 \sqrt{1-x^2} \sqrt{1-9x^2}} = \lim_{x \rightarrow 0} \frac{\sqrt{1-x^2} - \sqrt{1-9x^2}}{x^2} * \lim_{x \rightarrow 0} \frac{1}{\sqrt{1-x^2} \sqrt{1-9x^2}} =$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1-x^2} - \sqrt{1-9x^2}}{x^2} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{-\frac{x}{\sqrt{1-x^2}} + \frac{9x}{\sqrt{1-9x^2}}}{2x} = \lim_{x \rightarrow 0} \frac{9\sqrt{1-x^2} - \sqrt{1-9x^2}}{2\sqrt{1-x^2} \sqrt{1-9x^2}} = \frac{9-1}{2} = 4$$

Раскрытие неопределенности $0^0, \infty^0, 1^\infty$.

$$19) \quad \lim_{x \rightarrow 0+0} (\sin 2x)^{\frac{1}{\ln 3x}} = (0^0) = \lim_{x \rightarrow 0+0} \left(e^{\frac{\ln \sin 2x}{\ln 3x}}\right) = e^1 = e$$

$$\lim_{x \rightarrow 0+0} \frac{\ln \sin 2x}{\ln 3x} = \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow 0+0} \frac{\frac{1}{\sin 2x} \cos 2x * 2}{\frac{1}{x}} = \lim_{x \rightarrow 0+0} \frac{2x \cos 2x}{\sin 2x}$$

$$= \lim_{x \rightarrow 0+0} \frac{2x \cos 2x}{2x} = 1$$

$$20) \quad \lim_{x \rightarrow +\infty} (3^x + 2^x)^{\frac{1}{x}} = (\infty^0) = \lim_{x \rightarrow +\infty} e^{\frac{\ln(3^x + 2^x)}{x}} = e^{\ln 3} = 3$$

$$\begin{aligned}\lim_{x \rightarrow +\infty} \frac{\ln(3^x + 2^x)}{x} &= \left(\frac{\infty}{\infty}\right) = \lim_{x \rightarrow +\infty} \frac{\frac{1}{3^x + 2^x} (3^x \ln 3 + 2^x \ln 2)}{1} \\ &= \lim_{x \rightarrow +\infty} \frac{3^x \ln 3 + 2^x \ln 2}{3^x + 2^x} = \lim_{x \rightarrow +\infty} \frac{\ln 3 + \left(\frac{2}{3}\right)^x \ln 2}{1 + \left(\frac{2}{3}\right)^x} = \ln 3\end{aligned}$$

$$21) \quad \lim_{x \rightarrow 0} \left(\frac{2}{\pi} \arccos 2x\right)^{\frac{1}{\sin 5x}} = (1^\infty) = \lim_{x \rightarrow 0} \left(e^{\frac{\ln\left(\frac{2}{\pi} \arccos 2x\right)}{\sin 5x}}\right) = e^{-\frac{4}{5\pi}}$$

$$\lim_{x \rightarrow 0} \frac{\ln\left(\frac{2}{\pi} \arccos 2x\right)}{\sin 5x} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{-\frac{2}{\sqrt{1-4x^2}} \frac{1}{\arccos 2x}}{5 \cos 5x} = \frac{-2 \frac{2}{\pi}}{5} = -\frac{4}{5\pi}$$

$$22) \quad \lim_{x \rightarrow +\infty} \left(\frac{2}{\pi} \arctg 7x\right)^x = (1^\infty) = \lim_{x \rightarrow +\infty} \left(e^{x \ln\left(\frac{2}{\pi} \arctg 7x\right)}\right) = e^{-\frac{2}{7\pi}}$$

$$\begin{aligned}\lim_{x \rightarrow +\infty} \left(x \ln\left(\frac{2}{\pi} \arctg 7x\right)\right) &= (\infty * 0) = \lim_{x \rightarrow +\infty} \frac{\ln\left(\frac{2}{\pi} \arctg 7x\right)}{\frac{1}{x}} = \left(\frac{0}{0}\right) \\ &= \lim_{x \rightarrow +\infty} \frac{\frac{1}{\arctg 7x} \frac{7}{1+49x^2}}{-\frac{1}{x^2}} = \lim_{x \rightarrow +\infty} \frac{-7x^2}{(1+49x^2)\arctg 7x} = -\frac{1}{7} \frac{\pi}{2} \\ &= -\frac{2}{7\pi}\end{aligned}$$